

Answer 4 questions. You should indicate which 4 problems you wish to be graded. Write your solutions clearly in complete English sentences. You may quote theorems (within reason) as long as you state them clearly.

1. (10 points) In S_8 , write down a representative of each conjugacy class of elements of order 4 and compute the order of its centralizer.
2. (10 points) Let G be a group of order $124 = 2^2 \cdot 31$. Prove that G has a cyclic subgroup of order 62.
3. (10 points) Prove that there exists a nonabelian group of order $2025 = 3^4 \cdot 5^2$. (You do not have to determine its multiplication explicitly.)
4.
 - (a) (2 points) Give the definition of a maximal subgroup of a nontrivial group.
 - (b) (3 points) Find two maximal subgroups of S_5 of different orders.
 - (c) (5 points) Show that the group $(\mathbb{Q}, +)$ has no maximal subgroup.
5. Let G be a finite group and p a prime. Let K be the intersection of all the Sylow p -subgroups of G .
 - (a) (5 points) Prove that K is a characteristic subgroup of G .
 - (b) (5 points) Suppose H is a normal p -subgroup of G . Prove that $H \subseteq K$.
6. Let G be a group, and let $Z(G)$ denote its center.
 - (a) (4 points) State what it means for G to be nilpotent of class c .
 - (b) (1 point) Prove that if G is nilpotent of class 1, then G is abelian.
 - (c) (2 points) Define the commutator subgroup G' of G .
 - (d) (3 points) Prove that if G is nilpotent of class 2, then $G' \subseteq Z(G)$.